

TAMIL NADU STATE BOARD - STANDARD 12 PHYSICS VOLUME 1

CHAPTER 1: ELECTROSTATICS

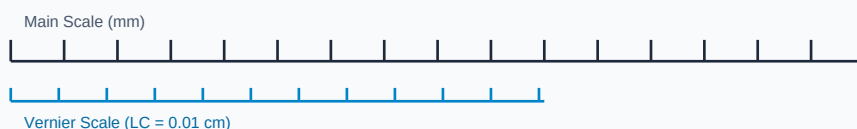
PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 12 English Medium
Subject:	Physics Volume 1
Chapter Name:	Electrostatics

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

Measurement Linear Scale Division

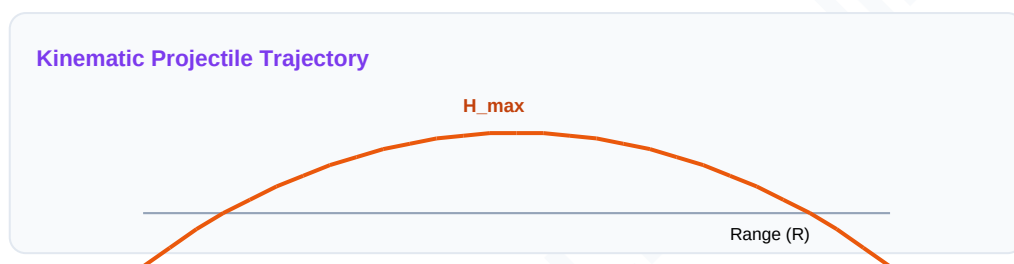


Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Coulomb Law	The electrostatic force between two point charges is proportional to the product of charges and inversely to distance squared.
Electric Field	The force experienced by a unit positive test charge placed at a point in space around a source charge.
Electric Dipole	A system of two equal and opposite point charges separated by a small distance.

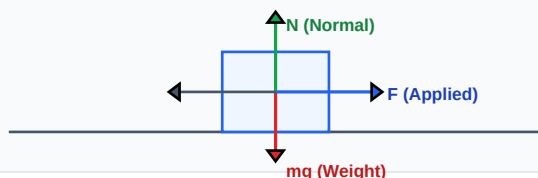
Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:

Force Vector Free Body Diagram (FBD)



Essential Formulas, Laws & Identities:

- Coulomb Force: $F = (1 / (4\pi\epsilon_0)) * (q_1 * q_2) / r^2$
- Electric Field: $E = F / q$
- Dipole Moment: $p = q * 2a$
- Electric Potential: $V = (1 / (4\pi\epsilon_0)) * q / r$
- Capacitance: $C = Q / V$ | Parallel Plate: $C = \epsilon_0 * A / d$

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

Review these standard board examination-style worked examples to learn proper step layout structures:

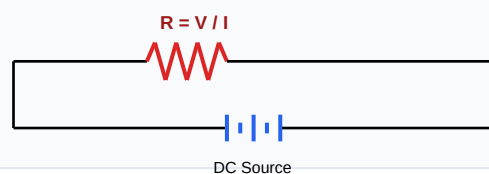
Example 1: Find the force between two charges of 2 micro-coulombs separated by 3 meters in vacuum.

Step-by-step Solution:

Apply Coulomb's law: $F = (9 \times 10^9) \times (2 \times 10^{-6} \times 2 \times 10^{-6}) / 3^2$.

$F = (9 \times 10^9) \times (4 \times 10^{-12}) / 9 = 4 \times 10^{-3}$ Newtons.

Schematic Electric Circuit Diagram



HOTS (Higher Order Thinking Skills) Challenge

Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

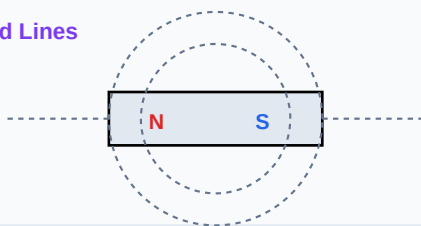
HOTS Challenge 1: Show that the electric field at an axial point of a dipole is twice that at an equatorial point at the same large distance.

Analytical Solution Strategy:

Axial Field: $E_{ax} = \frac{1}{4\pi\epsilon_0} \cdot \frac{2p}{r^3}$. Equatorial Field: $E_{eq} = \frac{1}{4\pi\epsilon_0} \cdot \frac{p}{r^3}$.

Comparing both yields $E_{ax} = 2 \cdot E_{eq}$, demonstrating axial fields drop less rapidly.

Magnetic Dipole Bar Field Lines



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

Q1 (1-Mark): State Gauss's Law in electrostatics.

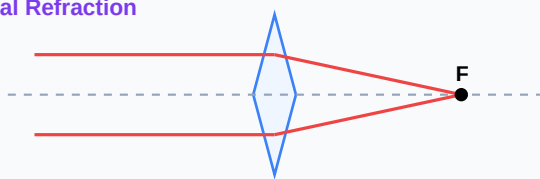
Q2 (2-Mark): What is an equipotential surface? State its properties.

Q3 (2-Mark): Calculate the energy stored in a 10 microfarad capacitor charged to 100V.

Q4 (5-Mark): Define dielectric constant of a medium.

Q5 (5-Mark): Explain the working principle of a Van de Graaff Generator.

Convex Lens Optical Refraction



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Inkjet Printers

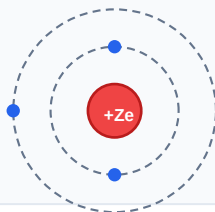
Deflection plates guide charged ink droplets to precise paper coordinates using electrostatic fields.

Application Case: Air Filter Systems

Electrostatic precipitators charge and trap smoke particles, reducing industrial factory pollution emissions.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

Bohr Atomic Shell Configuration



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3 , N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.
Good luck!

Sinusoidal AC Voltage Waveform

